RUI DING

Electrochemical Energy Devices & Chemical Sensors | Electrocatalysis & Membrane Electrode Assemblies | Closed-Loop Discovery of Material–Device Systems

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Research Areas: Electrochemical energy conversion devices, including proton-exchange-membrane electrolyzers and fuel cells; membrane electrode assemblies and catalyst-layer / ionomer interfaces; electrocatalyst discovery and synthesis for water splitting, oxygen reduction, and CO_2 reduction; active-learning-guided experimentation and closed-loop discovery; multi-fidelity computational validation (DFT/AIMD, machine-learned interatomic potentials); scientific agents and device digital twins for electrochemical cells and field-effect-transistor chemical sensors; and AI and workflow systems for automated experimentation

Citation: 1886 (2026-08-31); h-index: 24

ACADEMIC APPOINTMENTS

Eric and Wendy Schmidt AI in Science Fellow, *University of Chicago* — **Sep 2023 - Present**
Advisors: Prof. Junhong Chen (PME/Argonne) and Prof. Yuxin Chen (Computer Science).

Resident Associate, *Chemical Sciences and Engineering Division, Argonne National Laboratory* — **Sep 2023 - Present**
Host: Prof. Junhong Chen; research spanning AI/ML, nanomaterials, electrochemistry, sensors, and atomistic simulation.

EDUCATION

Ph.D. in Materials Science and Engineering, *Nanjing University, Nanjing, China* — **Sep 2018 - Jun 2023**
Research: machine learning, electrocatalysts, membrane-electrode assemblies, fuel cells, and water electrolyzers.

Visiting Graduate Researcher, *Department of Chemical and Biological Engineering, Hong Kong University of Science and Technology* — **Oct 2022 - Apr 2023**

Bachelor of Engineering in Renewable Energy Science and Engineering, *Nanjing University, Nanjing, China* — **Sep 2014 - Jun 2018**

SELECTED RESEARCH IMPACT

Active-learning-guided experimental materials discovery. Led a multistage ML workflow combining mining of 534 publications, five active-learning rounds with 258 synthesized multimetallic oxides, and domain-adapted DFT. The process reduced the failure rate (samples returning no usable overpotential at 10 mA cm^{-2}) from 47% to 0% and identified a Ru-Mn-Ca-Pr oxide that outperformed commercial IrOx at 1 A cm^{-2} and ran 125 h at 10/20 mA cm^{-2} in a PEM electrolyzer (Science Advances, eadr9038, 2025).

Autonomous in silico discovery for data-scarce materials and devices. Developed three complementary systems: T^3, which transforms scientific literature into device digital twins and enabled the screening of 123 million molecules (KDD 2026; Spotlight Oral, AI4Mat@ICLR 2026); DToR, which supports literature-grounded hypothesis generation across 27 materials and device topics (AI4Mat@NeurIPS 2025); and RAPIDS, an agent-accessible machine-learning-interatomic-potential (MLIP) scaffold for estimating and ranking probe–target non-covalent interaction energies (AI4Physics@ICML 2026).

Verifier-gated autonomous research and self-driving laboratory infrastructure. Built the Autonomous Research Compiler (ARC), a long-horizon in-silico closed loop that compiles literature-derived device states, routes candidate hypotheses through specialist scientific modules and executable physics checks, and revises the research plan when verification is adverse or inconclusive. ARC is hosted on the MADE-PUBLIC (NSF FMRG #2037026) data portal and cybermanufacturing infrastructure I designed and implemented, whose governed data, provenance, and modular services form its operating foundation; I am the technical lead for the AI/ML thrust of PoLARIS, the $20 million University of Chicago and Argonne node of the NSF Programmable Cloud Laboratories network, where a first control panel now sends workflow instructions to PolyBot, Argonne's self-driving laboratory, and monitors it in real time, and where I have built a prototype digital twin of the PolyBot cell; ARC is now running an autonomous campaign on a polymer system with the PolyBot team.

SELECTED DISTINCTIONS

Eric and Wendy Schmidt AI in Science Fellowship, Schmidt Sciences, Cohort 2 — **Sep 2023 - Present**

Spotlight Oral, AI4Mat at ICLR 2026 - Text-Twin-Translation (T^3) — **2026**

Industry AI research support

Anthropic Claude usage credits award, $20,000 (MIT–Anthropic AI for Science Study, 2026); Google Cloud Research Credits Program award, project #441314083982 (2026); OpenAI ChatGPT for Academic Researchers award, 12-month Pro-level research workspace for a five-member team (2026).

Journal cover features

Chemical Society Reviews (2024), Molecular Systems Design & Engineering (2025), Journal of Materials Chemistry A (2021), and Nano Research (2020).

PUBLICATIONS

*Selected first- and co-first-author publications and papers. 16 peer-reviewed in total. **Rui Ding** is bolded; * denotes equal contribution.*

Archival Conference Paper (full paper in archival proceedings)

[C1] **Rui Ding***, Zixin Ding*, Rodrigo P. Ferreira*, Yuxin Chen, and Junhong Chen. Text-Twin-Translation (T^3): A Full-Stack Machine Learning Framework for Functional Material-Device Systems Discovery. ***Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026), AI for Sciences Track. KDD is the flagship venue in data mining; acceptance rate 21%. Preliminary version selected for a Spotlight Oral at*** *AI4Mat@ICLR 2026*.

Peer-Reviewed Journal Articles

[J1] **Rui Ding*, Jianguo Liu*, Kang Hua, Xuebin Wang, Xiaoben Zhang, Minhua Shao, Yuxin Chen, and Junhong Chen. Leveraging Data Mining, Active Learning, and Domain Adaptation for Efficient Discovery of Advanced Oxygen Evolution Electrocatalysts. *Science Advances 11, eadr9038 (2025)***. doi:10.1126/sciadv.adr9038

[J2] **Rui Ding**, Junhong Chen, Yuxin Chen, Jianguo Liu, Yoshio Bando, and Xuebin Wang. Unlocking the Potential: Machine Learning Applications in Electrocatalyst Design for Electrochemical Hydrogen Energy Transformation. ***Chemical Society Reviews 53, 11390-11461 (2024)*** *(Journal cover)*. doi:10.1039/D4CS00844H

[J3] Rodrigo P. Ferreira*, **Rui Ding***, Fengxue Zhang, Haihui Pu, Claire Donnat, Yuxin Chen, and Junhong Chen. Expediting Field-Effect Transistor Chemical Sensor Design with Neuromorphic Spiking Graph Neural Networks. ***Molecular Systems Design & Engineering 10, 345-356 (2025)*** *(Journal cover)*. doi:10.1039/D4ME00203B

[J4] **Rui Ding**, Ran Wang, Yiqin Ding, Wenjuan Yin, Yide Liu, Jia Li, and Jianguo Liu. Designing AI-Aided Analysis and Prediction Models for Nonprecious Metal Electrocatalyst-Based Proton Exchange Membrane Fuel Cells. ***Angewandte Chemie International Edition 59, 19175-19183 (2020)***. doi:10.1002/anie.202006928

[J5] **Rui Ding**, Xuebin Wang, Aidong Tan, Jia Li, and Jianguo Liu. Unlocking New Insights for Electrocatalyst Design: A Unique Data Science Workflow Leveraging Internet-Sourced Big Data. ***ACS Catalysis 13, 13267-13281 (2023)***. doi:10.1021/acscatal.3c01914

[J6] **Rui Ding**, Yawen Chen, Ping Chen, Ran Wang, Jiankang Wang, Yiqin Ding, Wenjuan Yin, Yide Liu, Jia Li, and Jianguo Liu. Machine Learning-Guided Discovery of Underlying Decisive Factors and New Mechanisms for the Design of Nonprecious Metal Electrocatalysts. ***ACS Catalysis 11, 9798-9808 (2021)***. doi:10.1021/acscatal.1c01473

[J7] **Rui Ding**, Yiqin Ding, Hongyu Zhang, Ran Wang, Zihan Xu, Yide Liu, Wenjuan Yin, Jiankang Wang, Jia Li, and Jianguo Liu. Applying Machine Learning to Boost the Development of High-Performance Membrane Electrode Assembly for Proton Exchange Membrane Fuel Cells. ***Journal of Materials Chemistry A 9, 6841-6850 (2021)*** *(Inside cover)*. doi:10.1039/D0TA12571G

[J8] **Rui Ding**, Meng Ma, Yawen Chen, Xuebin Wang, Jia Li, Guoxiong Wang, and Jianguo Liu. Inspecting Design Rules of Metal-Nitrogen-Carbon Catalysts for Electrochemical CO2 Reduction Reaction: From a Data Science Perspective. ***Nano Research 16, 264-280 (2023)***. doi:10.1007/s12274-022-4763-y

[J9] **Rui Ding**, Yawen Chen, Zhiyan Rui, Kang Hua, Yongkang Wu, Xiaoke Li, Xiao Duan, Jia Li, Xuebin Wang, and Jianguo Liu. Machine Learning Utilized for the Development of Proton Exchange Membrane Electrolyzers. ***Journal of Power Sources 556, 232389 (2023)***. doi:10.1016/j.jpowsour.2022.232389

[J10] **Rui Ding**, Yawen Chen, Zhiyan Rui, Kang Hua, Yongkang Wu, Xiaoke Li, Xiao Duan, Xuebin Wang, Jia Li, and Jianguo Liu. Guiding the Optimization of Membrane Electrode Assembly in a Proton Exchange Membrane Water Electrolyzer by Machine Learning Modeling and Black-Box Interpretation. ***ACS Sustainable Chemistry & Engineering 10, 4561-4578 (2022)***. doi:10.1021/acssuschemeng.1c08522

[J11] **Rui Ding**, Shiqiao Zhang, Yawen Chen, Zhiyan Rui, Kang Hua, Yongkang Wu, Xiaoke Li, Xiao Duan, Xuebin Wang, Jia Li, and Jianguo Liu. Application of Machine Learning in Optimizing Proton Exchange Membrane Fuel Cells: A Review. ***Energy and AI 9, 100170 (2022)***. doi:10.1016/j.egyai.2022.100170

[J12] **Rui Ding***, Wenjuan Yin*, Gang Cheng, Yawen Chen, Jiankang Wang, Xuebin Wang, Min Han, Tianren Zhang, Yinliang Cao, Haimin Zhao, Shengping Wang, Jia Li, and Jianguo Liu. Effectively Increasing Pt Utilization Efficiency of the Membrane Electrode Assembly in Proton Exchange Membrane Fuel Cells through Multiparameter Optimization Guided by Machine Learning. ***ACS Applied Materials & Interfaces 14, 8010-8024 (2022)***. doi:10.1021/acsami.1c23221

[J13] **Rui Ding**, Yawen Chen, Xiaoke Li, Zhiyan Rui, Kang Hua, Yongkang Wu, Xiao Duan, Xuebin Wang, Jia Li, and Jianguo Liu. Atomically Dispersed, Low-Coordinate Co-N Sites on Carbon Nanotubes as Inexpensive and Efficient Electrocatalysts for Hydrogen Evolution. ***Small 18, 2105335 (2022)***. doi:10.1002/smll.202105335

[J14] **Rui Ding***, Wenjuan Yin*, Gang Cheng, Yawen Chen, Jiankang Wang, Ran Wang, Zhiyan Rui, Jia Li, and Jianguo Liu. Boosting the Optimization of Membrane Electrode Assembly in Proton Exchange Membrane Fuel Cells Guided by Explainable Artificial Intelligence. ***Energy and AI 5, 100098 (2021)***. doi:10.1016/j.egyai.2021.100098

[J15] **Rui Ding**, Yide Liu, Zhiyan Rui, Jia Li, Jianguo Liu, and Zhigang Zou. Facile Grafting Strategy Synthesis of Single-Atom Electrocatalyst with Enhanced ORR Performance. ***Nano Research 13, 1519-1526 (2020)*** *(Back cover)*. doi:10.1007/s12274-020-2768-y

Preprints and Peer-Reviewed Workshop Papers

[W1] **Rui Ding***, Zixin Ding*, Rodrigo P. Ferreira*, Yuxin Chen, and Junhong Chen. Geometry, Not Energy Surface, Drives the Neutral MLIP-DFT Gap in Atomistic Interaction Surrogates. ***AI4Physics, at ICML 2026***

[W2] **Rui Ding**, Rodrigo P. Ferreira, Yuxin Chen, and Junhong Chen. Hierarchical Deep Research with Local-Web RAG: Toward Automated System-Level Materials Discovery. ***arXiv preprint (2025); preliminary version at AI4Mat, at NeurIPS 2025***. arXiv:2511.18303

[W3] Rodrigo P. Ferreira*, **Rui Ding***, Rapti Ghosh, Haihui Pu, and Junhong Chen. Neuromorphic Random Walk for Experimental Phosphate Adsorption Modeling. ***Machine Learning and the Physics Sciences (ML4PS), at NeurIPS 2025***

[W4] **Rui Ding**, Shira M. Lupkin, and Nicole Tebaldi. 1 in 4 Committed Links, 1 in 34 Mentions: When Aggregate Metrics Hide Canonicalization Failures in Clinical Knowledge Graphs. ***GenAI4Health 2026 (Main Track)***

[W5] Rodrigo P. Ferreira, Ruijie Zhu, **Rui Ding**, Haihui Pu, Yuxin Chen, and Junhong Chen. PALM: Physics-Aware Leakage Minimizer. ***Machine Learning and the Physical Sciences (ML4PS), at NeurIPS 2026***

Manuscripts Under Review

[U1] **Rui Ding**, Haihui Pu, Kuldip Kumar, Jonathan S. Grabowy, Dominic A. Brose, Maria Chan, Seth B. Darling, Andrew Ferguson, Yuxin Chen, and Junhong Chen. From Laboratory Promise to Field Readiness: AI/ML for Discovering Water Materials That Survive Real Water. ***Under review at Nature Water***

[U2] **Rui Ding**, Peng Gong, Rodrigo P. Ferreira, Yuxin Chen, and Junhong Chen. Hierarchical Deep Research with Local-Web RAG: Toward Automated System-Level Materials Discovery. ***Under review at Nature Machine Intelligence*** (Extended manuscript based on W2)

[U3] **Rui Ding**, Zixin Ding, Rodrigo P. Ferreira, Yuxin Chen, and Junhong Chen. Reliability-Aware Routing for Foundation Atomistic Simulators in Non-Covalent Binding. ***Under review at AAAI 2027*** (Extended manuscript based on W1)

[U4] Haihui Pu*, **Rui Ding***, Tetyana Ignatova, Eric A. Josephs, Yuxin Chen, Wei Chen, and Junhong Chen. Nano-AI Convergence: Accelerating Materials Discovery and Overcoming Device Variability in Nanotechnology ***Under review at Digital Discovery***

[U5] **Rui Ding** et al. Compiling Literature into Device States for Verifier-Gated Closed-Loop Design. *Submitted to AI4Mat at NeurIPS 2026 (under review).*

RESEARCH SUPPORT AND COMPUTING AWARDS

Research Support

AI in Science Schmidt Fellowship Seed Fund ($33,000) - **Principal Investigator**, *University of Chicago* **2026**
AI-Guided Discovery of PFAS-Selective Membrane Interfaces: From Autonomous Design to Experimental Validation.

Science Labs that Only AI Can Build, - **ML Lead** *University of Chicago AI Research Area* **2025**
Text-Twin-Translation: AI-guided discovery of robust PFAS chemical FET sensors and bio-FET platforms in real matrices.

NSF Supplemental Award - FMRG 2037026 and DMREF 2323296, - **ML Lead** *National Science Foundation* **2025**
ML-driven integration of 2D materials for printable sensors in biosensing and environmental monitoring.

AI+Science Research Initiative ($5,000) - **Principal Investigator** *Data Science Institute, University of Chicago* **2024**
BRAINIAC: AI-driven optimization of nano-micro materials and devices.

External Computing and User Facility Awards

Allocations and facility awards: NERSC Exploratory Allocation ERCAP0039011 / m5358 (2026); NSF ACCESS MAT240097 (2024-2026); Center for Nanoscale Materials Service Unit Award (2024); Argonne Leadership Computing Facility ID16430 (2024).

RESEARCH SOFTWARE AND SCIENTIFIC INFRASTRUCTURE

MADE-PUBLIC Data Portal and Cybermanufacturing Platform (NSF FMRG #2037026) - Lead designer/developer. Integrated Globus data management, searchable metadata, interactive knowledge graphs, Bayesian experiment design, sensor screening, atomistic simulation, and manufacturing blueprints in a browser-accessible platform. **2024 - Present**

Deep Tree of Research (DToR) – Project lead and first author. Local-first, tree-structured agent for long-horizon synthesis, gap-driven branching, provenance-rich reporting, and materials hypothesis generation. **2025 - Present**

Text-Twin-Translation (T^3) – Project lead and first author. Literature extraction, topology-aware device digital twins, and component-level candidate screening for complex functional devices. **2025 - Present**

RAPIDS - Project lead and first author. Guarded multi-fidelity MLIP scaffold with pose search and chemistry-aware escalation to DFT. **2026 - Present**

Autonomous Research Compiler (ARC) / PoLARIS - ARC lead architect/developer and first author; technical lead for the AI/ML thrust of PoLARIS. Built a verifier-gated, long-horizon in-silico research loop on the MADE-PUBLIC foundation, integrating literature-derived device states, specialist scientific modules, durable Globus state, and fail-closed physics checks. A first control panel now sends workflow instructions to PolyBot, Argonne's self-driving laboratory, and monitors it in real time; built a prototype digital twin of the PolyBot cell; ARC is running an autonomous campaign on a polymer system with the PolyBot team. **2026 - Present**

Open-source research code. Maintains public repositories spanning electrocatalyst data mining, device optimization, FET sensing, scientific agents, and AI-for-materials workflows.

TEACHING, MENTORING, AND EDUCATIONAL LEADERSHIP

Faculty Research Mentor - Eric and Wendy Schmidt AI in Science Faculty Fellowship; mentored Prof. Omolola Ogbolumani (University of Lagos) on human-in-the-loop AI for food-energy-water nexus optimization. **Jan - Jun 2026**

Curriculum Developer for NSF Research Traineeship (NRT), University of Chicago. Developed and open-sourced eight self-contained AI/ML teaching notebooks for Ph.D. students in the Pritzker School of Molecular Engineering. **2026**

Technical Lead - University of Chicago AI+Science Hackathon, multi-agent framework track; supplied source code and datasets. **2026**

Research Mentor - RENEU program; mentored Covenant Amoo (Obafemi Awolowo University; Chemical Engineering; Undergraduate Student; Remote) on LLM-based knowledge extraction for FET sensors. **2025**

Research Mentor - mentored Maya Martinez (University of Chicago; Pritzker School of Molecular Engineering; Undergraduate Student) on MD simulation for PFAS contaminant adsorption behavior on Ni-HHB/Ni-HHTP MOFs.

Research Mentor - mentored Samuel Effendy (University of Illinois Chicago; Computer Science; Undergraduate Student) on data portal development

Assistant Research Mentor - RENEU program; assisted in mentoring Abdulafeez Olaitan (University of Lagos; Applied Physics; Undergraduate Student; Remote) on *Dynamic Reshaping of Nanodroplet Under Electron Beam Radiation*. **2024**

Undergraduate Team Mentor - University of Chicago AI+Science Hackathon; guided a four-student team on AI-enabled materials characterization. **2025**

Host Representative - AI+Science Schmidt Fellows Speaker Series, University of Chicago. **2023 - 2026**

Courses prepared to teach or develop: Chemical engineering core: thermodynamics, transport phenomena, chemical reaction engineering, modeling and simulation, and the unit-operations laboratory. Electives: electrochemical engineering and electrochemical energy systems; catalysis and interfacial engineering; data science and machine learning for chemical engineering; nanomaterials

SELECTED PRESENTATIONS

2027 Compiling Literature into Device States for Verifier-Gated Closed-Loop Design *[Poster, accepted]*. SLAS2027 International Conference and Exhibition, San Diego, CA.

2026 Autonomous Research Compiler: An AI Researcher for Functional Materials and Devices *[Poster, accepted]*. 2026 MRS Fall Meeting, Symposium MT01: Leveraging Advances in AI for Materials Design and Autonomous Materials Science, Boston, MA (December 2, 2026).

2026 Text-Twin-Translation (T^3): A Full-Stack Machine Learning Framework for Functional Material-Device Systems Discovery ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), Jeju Island, Republic of Korea.

2026 Text-Twin-Translation (T^3): A Full-Stack Machine Learning Framework for Functional Material-Device Systems Discovery, preliminary version [Spotlight Oral]. AI4Mat at ICLR 2026, Brazil.

2026 PoLARIS platform demonstration video [Project Demonstration]. NSF National Network of AI-Programmable Cloud Laboratories Summit (PCL Kickoff), Washington, D.C.

2026 Geometry, Not Energy Surface, Drives the Neutral MLIP-DFT Gap in Atomistic Interaction Surrogates *[Poster]*. AI4Physics at ICML 2026, Seoul, Republic of Korea.

2025 Hierarchical Deep Research with Local-Web RAG: Toward Automated System-Level Materials Discovery *[Poster]*. AI4Mat at NeurIPS 2025, San Diego, CA.

2025 Neuromorphic Random Walk for Experimental Phosphate Adsorption Modeling. *[Poster]*. ML4PS at NeurIPS 2025, San Diego, CA.

2025 Application of Versatile Machine Learning Approaches to Facilitate the Discovery of Nanomaterials and Devices in Renewable Energy *[Contributed Talk]*. Gordon Research Seminar (GRS) on Nanomaterials for Applications in Energy Technology, Ventura, CA.

2025 Journey to a Nanomaterials/Devices Copilot *[Invited Presentation]*. Knowledge-Guided Machine Learning Workshop, University of Michigan, Ann Arbor, MI.

2024-2025 Cybermanufacturing Platform of the MADE-PUBLIC Project *[Project Presentation]*. NSF MADE-PUBLIC Annual Retreat, Northwestern University, Evanston, IL & Iowa State University, Ames, IA

PROFESSIONAL SERVICE

Community standards: Co-author, Transparent reporting for agentic catalysis enabled by artificial intelligence (TRACE-AI): Community guidelines and a publication checklist. Chem Catalysis 6, 101755 (2026). doi:10.1016/j.checat.2026.101755

Conference and workshop reviewer: ACM SIGKDD Main Conference; NeurIPS AI4Mat; ICLR AI4Mat; ICML AI4Physics.

Journal reviewer: ACS Catalysis; Energy & Fuels; Journal of Water Process Engineering; Synthetic Metals; Organic Letters; ACS Applied Energy Materials; Nanomaterials.

TECHNICAL EXPERTISE

AI/ML & agents: active learning, Bayesian optimization, graph/spiking/equivariant ML, uncertainty-aware and multi-fidelity learning, explainable AI, RAG, local/open-source LLMs, LLM fine-tuning, agent harness, prompt optimization and fine-tuning, agent orchestration, scientific text mining, provenance, and tool use; **software:** Python, R, Linux, Git, Django/Flask, HTML/CSS/JavaScript, Globus, HPC workflows, and visualization.

Computational/experimental materials: DFT/AIMD (VASP, QE), ML interatomic potentials, molecular dynamics (GROMACS), Gaussian/ORCA, and COMSOL; nanomaterial synthesis, electrochemistry, characterization, fabrication, and field-effect-transistor (FET) sensing.